



Wollo University
Kombolcha Institute of Technology
Department of Computer Science

Course Information

Course Title	Web Programming				
Course Code	CoSc 3081				
ECTS	7				
Contact Hours	Lecture	Lab/Practical	Tutorial	Home Study	Total
	48	48	16	55	165
Prerequisite	None		Target Group	2 nd year CS students	
Academic Year	2018 E.C		Lecture days and Hours		
Semester	I		Class Room	Lab-1052-2	

Instructor Information

Instructor Name	Ramizer Arega	email	Ramizer.arega@kiot.edu.et
Office [consultation] Hours		Office	Bank building Off_No:203

Course description

Client-server Architecture: Web page Design and development; information Architecture & visualization; static & dynamic pages, client side programming using scripting languages Client-server Architecture: Web page Design and development; information Architecture & visualization; static & dynamic pages, client side programming using scripting languages.

Learning Outcomes [Competency]

- The concepts, principles and methods in programming for web and Internet environment and the services and technologies available and used on Internet. Specifically, at the completion of this module the student is able to
- Provide fundamental concepts and skills for the understanding and development of web- based applications.
- Construct Web sites using HTML and Server Side scripting Languages.
- Implement client-side interactivity
- Use CSS to manage Web site presentation and maintenance
- Select and customize existing technologies to set up and maintain web servers
- Specify, build and manage form and content of information-rich web sites
- Design, implement and evaluate client-server systems following specific protocol specifications, taking into account concurrency issue.

Course contents

Chapter	Title	Content
1	The world Wide Web	1.1. Course overview 1.2. The Internet and World Wide Web 1.3. Web Hosting and Domain Name Registration
2	Hyper Text Markup Language (HTML)	2.1 Introduction to HTML 2.2. HTML tags 2.2.1. Basic HTML tags (HTML, HEAD, BODY, TITLE) 2.2.2. Meta tag 2.2.3. HTML Comments 2.2.4. HTML Link 2.2.4. HTML Text Formatting tags 2.5. HTML image inserting tag 2.2.6. HTML Table 2.2.7. Ordered and Unordered List in HTML 2.2.8. HTML Frames 2.2.8.1. Frame Set 2.2.8.2. Internal Frame 2.2.9. HTML Form and Form Controls 2.2.9. Inserting Multimedia in HTML 2.2.9.1. Embed vs Video and Audio tags 2.2.10. HTML Graphics 2.2.10.1. HTML Canvas 2.2.10.2. HTML SVG
<u>3</u>	Cascading Style Sheet (CSS)	3.1. CSS Basics 3.2. Style Sheet Rules 3.3. Style Properties 3.4. CSS Measuring Units
4	Client Side Scripting (JavaScript)	4.1. Introduction to JavaScript 4.2. JavaScript Basic 4.3. JavaScript Comments 4.4. Basic JavaScript Input Output 4.5. JavaScript Data Types and Variables 4.6. Arithmetic and Logical Operators in JavaScript 4.7. Control Structures (Conditional and Looping Statements) 4.8. Array in JavaScript 4.9. JavaScript Functions 4.10. JavaScript DOM (Document object Model) 4.11. Form Processing using JavaScript 4.12. JavaScript BOM (Browser Object Model)

<u>5</u>	Chapter 5: Server-Side Scripting (PHP)	5.1 Mechanics of Creating a Program (C++) 5.2 Compiling Programs 5.3 IDE Tools 5.4 The Structure of C++ Programs 5.5 Comments 5.6 C++ Basic Input/Output 5.7 Constants, Keywords, Variables, Data Types and Operators: 5.8 Operators Precedence in C++ 5.9 Simple Type Conversion 5.10 Statements: Null Statement, The Block Statements, The Assignment Statements. 5.11. PHP File Upload 5.12. PHP Cookies and Session 5.13. Database Programming using PHP 5.14. PHP File Input-Output 5.15. PHP Date and Time 5.16. PHP Mathematical Functions 5.17. PHP OOP
<u>6</u>	Advanced JavaScript and XML (AJAX)	6.1. Introduction to AJAX 6.2. XMLHttpRequest Object 6.3. Sending Request to PHP server 6.4. Handling Response from Server
<u>7</u>	Introduction to web development frameworks	a. Bootstrap & jQuery b. Node.js c. Angular.js d. React.js

Course Assessment Methods

Continuous Assessment Method		Weight
Quiz assignment project		30%
Mid exam	Test I	20 %
Final exam	Test II	50%
Total		100%

Reference

- Fundamental Concepts for Web Development: HTML5, CSS3, JavaScript and much more!, by Roxane Anquetil
- Web Design with HTML, CSS, JavaScript and jQuery Set 1st Edition by Jon Duckett (Author) W3Schools, <https://www.w3schools.com> .